

The Carry-Trade Tournament

What the game was testing

FINA3020 International Finance

1. What the game asked you to do

You ran a foreign exchange funding desk with \$100,000 of equity. The game had five stages.

1. You predicted whether uncovered interest parity holds, or whether the carry trade earns a systematic excess return.
2. You borrowed Japanese yen at 1.00% and invested the proceeds across three higher-yielding currencies: the US dollar at 3.75%, the Brazilian real at 14.00%, and the Mexican peso at 6.50%. You also chose gross leverage, between one and five times equity.
3. Six months in, funding conditions tightened. The VIX rose from 14 to 38, the Bank of Japan raised its policy rate by 50 basis points, and you were shown a stress scenario for the remaining six months: the real down 14%, the peso down 11%, and the yen up 12.5% against the dollar. You chose to de-leverage, to hold, or to borrow more.
4. You saw your final equity, and a slider that asked how likely you thought the stress scenario was.
5. You wrote a one-sentence reflection.

The game was not a forecasting contest, and the score was never the point. It was built to make one thing concrete: a carry trade pays you a small, steady, predictable amount, and charges you an occasional large, sudden, unpredictable one. Everything below is what that sentence means in numbers.

2. The formula behind every number in the game

This is the payoff identity from lecture. Consider a one-year carry trade funded in currency F (low yield) and invested in currency H (high yield), where S is units of F per unit of H :

$$\frac{E_1}{E_0} = L \left[(1 + i_H) \frac{S_1}{S_0} - (1 + i_F) \right] + (1 + i_F), \quad \text{ROI} = \frac{E_1}{E_0} - 1$$

Here E_0 and E_1 are equity at the start and end of the year, L is gross leverage (assets divided by equity), so borrowing is $(L - 1)E_0$, and i_H and i_F are the two interest rates. A value of S_1/S_0 below one means the funding currency has appreciated, which is the loss case.

That formula measures the return *in the funding currency*. The game reported everything in US dollars, so the same accounting reads

$$\frac{E_1}{E_0} = L \sum_k w_k (1 + i_k) \frac{X_{1,k}}{X_{0,k}} - (L - 1)(1 + i_{\text{JPY}}) \frac{S_0}{S_1},$$

where w_k is the weight of currency k , X is the US dollar value of one unit of that currency, and S_0/S_1 is how much more US dollars it takes to repay each yen of debt. The dollar leg and any uninvested cash have $X_1/X_0 = 1$; cash is simply a leg with $i_k = 0$.

Take the portfolio the game opened with: three times leverage, 40% dollars, 30% reals, 30% pesos. With no currency movement at all,

$$\frac{E_1}{E_0} = 3.00 \times [0.40(1.0375) + 0.30(1.1400) + 0.30(1.0650)] - 2.00 \times (1.0100) = 1.2095,$$

an ROI of +20.95%. Under the Stage 3 scenario the currency ratios stop being one. The yen multiplier becomes 1.1250, because the dollar/yen rate falls from 150.00 to 133.33 and each yen of debt therefore costs 12.5% more dollars to repay, and the real and peso legs are multiplied by 0.86 and 0.89:

$$\begin{aligned} \frac{E_1}{E_0} &= 3.00 \times [0.40(1.0375)(1.0000) + 0.30(1.1400)(0.8600) + 0.30(1.0650)(0.8900)] \\ &\quad - 2.00 \times (1.0100)(1.1250) = 0.7079, \end{aligned}$$

an ROI of −29.21%. Nothing else in the game is doing anything more complicated than this.

3. Where a carry return actually comes from

Read the dollar formula as two pieces.

The first piece is the interest spread. It is positive, it is known the day you put the trade on, and it accrues steadily. It is the reason anyone does this.

The second piece is the currency movement. It is unknown, and it is volatile. It enters multiplicatively on both the asset side and the debt side.

Leverage, L , multiplies both pieces. This is the part that is easy to miss when the only number on your screen is a positive expected return. Going from three times to five times leverage on the reals raised the no-movement ROI from 20.95% to 66.00%. It also raised the loss from a one-point adverse currency move from \$4,005 to \$9,740, and it moved the point of total loss from a 30-point move to a 17-point move. The reward grew by a factor of about three; the fragility grew faster.

4. How to size the risk before you trade

This is the arithmetic the game wanted you to do at Stage 2, before you locked anything in. It takes about a minute and it is exact, not an approximation.

Call one **point** a single percentage point of simultaneous adverse movement: the emerging market currencies each fall 1% against the dollar while the yen gains 1% against the dollar. Write A_{safe} for the year-end value of the legs with no currency exposure (the dollar leg and cash), A_{EM} for the year-end value of the real and peso legs before any currency movement, and D for the year-end yen debt valued at the starting exchange rate. Then year-end equity under a move of x points is

$$E_1 = \underbrace{(A_{\text{safe}} + A_{\text{EM}} - D)}_{\text{quiet-market cushion}} - x \cdot \underbrace{0.01(A_{\text{EM}} + D)}_{\text{loss per point}}.$$

Equity is *linear* in the size of the move, so two numbers describe the whole risk, and the distance to total loss follows immediately:

$$\text{Loss per point} = 1\% \times (A_{\text{EM}} + D)$$

$$\text{Distance to wipeout} = \text{quiet-market cushion} \div \text{loss per point}$$

For the opening portfolio: $A_{\text{EM}} = \$198,450$ and $D = \$202,000$, so each point costs \$4,005, or 4.00% of equity. The cushion is \$120,950. Dividing, the desk is wiped out by a 30.2-point move. Here are three portfolios you could have built, all with the same \$100,000 of equity.

	Cautious 2×, 70 USD / 30 MXN	Opening 3×, 40/30/30	Maximum carry 5×, 100% BRL
ROI if nothing moves	+8.15%	+20.95%	+66.00%
Loss per point	−1.65%	−4.00%	−9.74%
Distance to wipeout	65.6 pts	30.2 pts	17.0 pts
Real and peso crash alone	no wipeout	−60.9%	−29.1%
Yen rally alone	+107.1%	+59.9%	+41.1%

For scale, in the yen carry unwind of August 2024 the dollar/yen rate fell roughly 12% in about a month, and over the second half of 2008 it fell roughly 20%. Those are moves in the funding currency alone, before anything happens to emerging market assets. A desk running the maximum-carry portfolio is inside that range.

One warning about stress tests. A single scenario is not a risk measure. Run a moderate test on these portfolios, the emerging market currencies down 8% and the yen up 6%, and the opening portfolio returns −7.05% while the maximum-carry portfolio returns only −3.84%. The riskier portfolio looks *safer*, because a 14% Brazilian yield absorbs an 8% depreciation. The moderate test simply does not reach far enough out to find where the maximum-carry portfolio breaks. The distance to wipeout, 17.0 points against 30.2, tells you the truth that the single scenario hides.

Finally, note that none of this models a margin call. A real desk at 17 points of distance would be closed out by its prime broker long before equity actually reached zero, at prices it did not choose.

5. Why the Stage 3 shock hit both sides at once

The scenario was not an arbitrary run of bad luck. It was built to reproduce the way carry trades actually unwind.

In calm markets, investors borrow the low-yield currency and buy the high-yield one, which pushes the funding currency down and the target currencies up. The trade pays. When funding conditions tighten, everyone reverses at once: positions are closed, the funding currency is bought back, and the high-yield currencies are sold. The funding currency appreciates *because* the risky assets are being sold, not independently of it.

That is why the game moved the yen up at the same moment it moved the real and peso down. Both sides of your balance sheet lose together, and the diversification you thought you had across three target currencies does not help, because the yen leg is common to all of them. This is the correlation, not the volatility, that makes the trade dangerous.

6. The three responses, and why none is right in advance

For the opening portfolio, this is what each response produced.

Response	Final equity	ROI	Year-end yen debt
A. De-leverage	\$90,990	−9.01%	\$120,281
B. Hold	\$70,793	−29.21%	\$227,250
C. Increase borrowing	\$49,405	−50.59%	\$340,594

At five times leverage in reals the same three choices gave −6.97%, −64.30% and −107.08%. The last figure is below −100%: the debt exceeded the assets, and the desk did not merely lose its capital, it ended the year owing money.

It is tempting to read these tables as saying that de-leveraging is correct. It is not. De-leveraging is the best of the three *in this scenario*, and it would have been the worst of the three had the sell-off reversed, because it sells assets at the depressed month-six prices, those assets do not recover if prices recover, and it surrenders the remaining carry on everything it sold. Holding keeps the full spread and the full exposure. Borrowing more is a deliberate bet that the market has overreacted, and it raises leverage precisely when the loss per point is already elevated.

The useful conclusion is the one a trading desk lives by: you cannot fix a position size after the news arrives. Every response available at Stage 3 was a choice between realising a loss and enlarging an exposure. The decision that actually determined your outcome was the leverage you chose at Stage 2, when the screen was calm and the only number in front of you was positive.

7. What the probability slider was asking

Stage 4 asked you to assign a probability p to the stress scenario and then computed

$$\mathbb{E}[\text{ROI}] = (1 - p) \text{ROI}_{\text{quiet}} + p \text{ROI}_{\text{stress}}, \quad p^* = \frac{\text{ROI}_{\text{quiet}}}{\text{ROI}_{\text{quiet}} - \text{ROI}_{\text{stress}}},$$

where p^* is the crash probability at which the expected return falls to zero. For the opening portfolio held through the shock, p^* is about 42%: a crash would have to arrive in roughly two years out of five before this trade stopped being worth doing on expectation.

That number is the bridge back to the prediction you made in Stage 1.

The forward premium puzzle. Uncovered interest parity says high-yield currencies should depreciate by enough to erase the interest differential. Historically they have not, and carry trades have earned positive average returns. This is a genuine and well-documented empirical failure.

The peso problem. Average returns measured over a sample that happens to contain no disaster will overstate what you should expect, because the disaster is in the distribution even when it is not in the data. Your moderate stress test in Section 4 is this problem in miniature: it was calibrated to an ordinary bad week and it understated the tail badly.

The crash-risk premium. Carry returns are negatively skewed. Many small gains are interrupted by occasional large losses, and the losses arrive in exactly the states of the world where investors can least afford them, when funding is scarce and other assets are also falling. Part of the carry premium is therefore not a free lunch at all. It is payment for accepting a loss precisely when a loss hurts most.

So a positive average carry return is not by itself evidence that uncovered interest parity fails in an exploitable way. Estimating how much of it is genuine mispricing and how much is compensation for the tail is an open research question, and it is the reason the game asked you for p rather than telling you what p was.

8. Five things to take away

1. A carry return is an interest spread plus a currency movement, and leverage multiplies both. The spread is what you are paid; the currency movement is what you are paid *for*.
2. Two numbers size the whole risk before you trade: the loss per point of adverse movement, and the quiet-market cushion divided by it, which is your distance to wipeout.
3. A single stress scenario can make a more dangerous portfolio look safer. Check the distance to wipeout, not one scenario.
4. Carry trades unwind with the funding currency and the target currencies moving against you together. That correlation is the risk, and holding several target currencies does not diversify it away.
5. The position size chosen in calm markets determines the outcome. By the time the news arrives, every remaining option is already expensive.